

DigiPermit — Major Project System Description

Project: DigiPermit — Multi-User Foreign-National Visa and Permit Compliance Monitoring System

Module integration: DS3 × IS3 × IP2

Document version: 1.0

Date: June 2026

Repository: <https://github.com/MrNtuli/DPermit>

Live application: <https://mrntuli.github.io/DPermit/>

Live API: <https://dpermit.onrender.com>

Important limitation

DigiPermit is a **compliance-monitoring and verification platform**. It does **not** issue official visas or immigration permits and does **not** replace the Department of Home Affairs or any official immigration authority. Organisations capture, monitor, verify, and audit compliance records only.

Table of contents

1. [Executive summary](#)
2. [Problem and solution](#)
3. [What the system does — end-to-end](#)
4. [How DigiPermit meets Major Project specifications](#)
5. [System architecture](#)
6. [Process diagrams \(BPMN-style\)](#)
7. [User roles and what happens for each](#)
8. [Verification engine](#)
9. [Data model \(IS3\)](#)
10. [Analytics and reporting \(IS3\)](#)
11. [IoT simulation \(IS3\)](#)
12. [API layer \(IP2\)](#)
13. [Agile delivery \(DS3\)](#)
14. [Testing and quality assurance](#)
15. [Deployment architecture](#)
16. [Known limitations and future work](#)

1. Executive summary

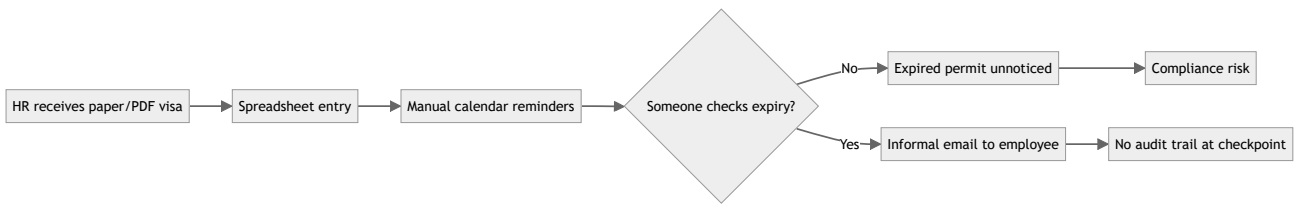
DigiPermit is a full-stack web platform that helps **employers, universities, clinics, foreign nationals, verification officers, and compliance managers** track immigration-related permit records, receive expiry warnings, verify documents at checkpoints, and maintain structured audit trails.

The system was delivered using **Agile Scrum** over four sprints, integrating:

Module	Lecturer focus	DigiPermit deliverable
IS3	BPMN, 3NF database, IoT map, analytics, AI workflow	PostgreSQL schema, process models, SQL views, Power BI plan, rule-based insights
IP2	REST API, integration, testing	Express API (40+ endpoints), Postman collection, Ionic frontend
DS3	Agile process, documentation, GitHub	Sprint reviews, backlog, repository, demo script, test guide

2. Problem and solution

2.1 As-Is problem (manual compliance)



Pain points: missed expiry dates, no real-time verification, weak audit trails, uncontrolled access to sensitive data, manual reporting.

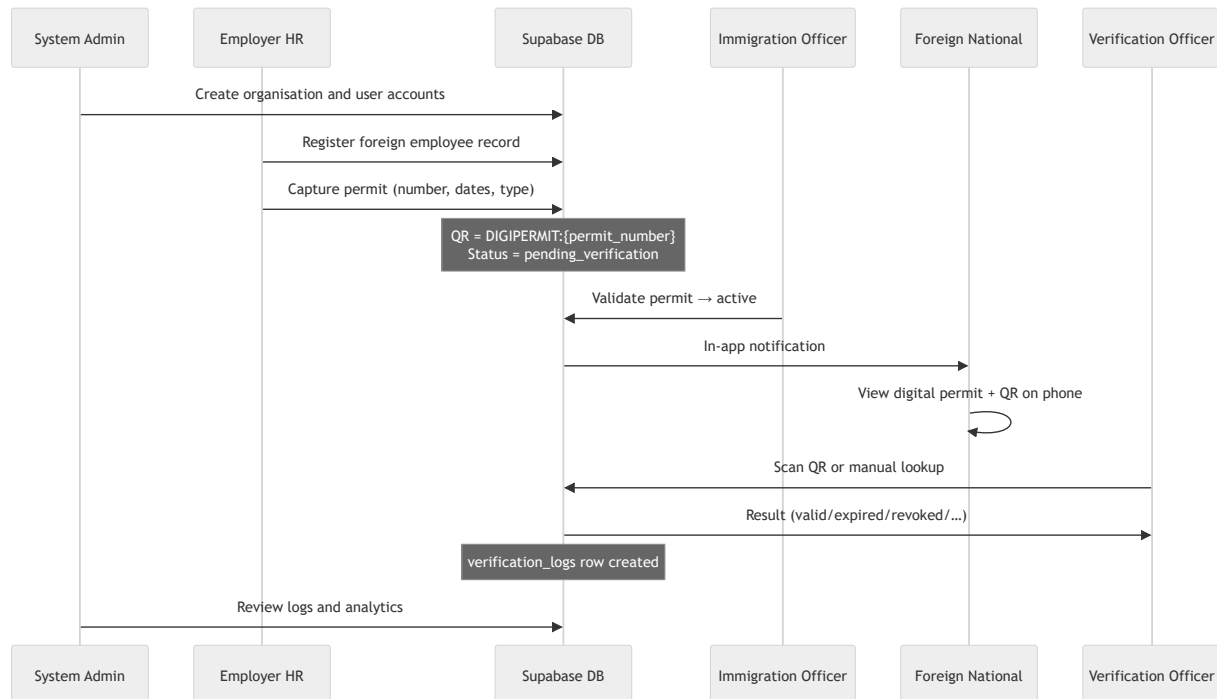
2.2 To-Be solution (DigiPermit)



Outcome: centralised records, automated expiry monitoring, multi-channel verification, immutable logs, role-based access, filterable analytics.

3. What the system does — end-to-end

3.1 Typical lifecycle (work visa at an employer)



3.2 What happens when a permit is captured

1. HR (or university/clinic officer) selects an existing **foreign national** record.
2. HR enters permit number, type, issue/expiry dates, passport number.
3. Backend generates:
 - `qr_code_value` = `DIGIPERMIT:{permit_number}`
 - `rfid_tag` (simulated identifier)
 - Initial status `pending_verification`
4. Immigration officer (simulated role) **validates** or **rejects** the record.
5. On validation, status becomes **active** (or **expiring_soon** if near expiry).
6. **Daily cron job** recalculates statuses and creates **notifications** and **alerts** at configured thresholds (90, 60, 30, 14, 7, 1, 0 days).

3.3 What happens at a checkpoint

1. Verifier opens **Manual Lookup**, **QR Scan**, or **RFID Simulation**.
 2. System loads permit from database and evaluates:
 - Exists? Revoked? Expired? Expiring soon?
 - QR/RFID match?
 3. Returns one of **ten verification result states** (e.g. valid, expired, revoked, not_found, suspicious).
 4. Writes row to `verification_logs` ; may create `alerts` for failed/suspicious scans.
 5. Frontend shows **verification certificate** (PNG/PDF/print).
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4. How DigiPermit meets Major Project specifications

4.1 IS3 — Information Systems 3

IS3 requirement	How DigiPermit addresses it	Evidence
BPMN As-Is / To-Be process models	Manual spreadsheet process vs automated DigiPermit workflow	IS3-BPMN-PROCESS-MODELS.md , Section 6 below
3NF relational database	12 normalised tables, FK constraints, no transitive dependencies	IS3-ERD.md , database/migrations/
Row Level Security	RLS policies on Supabase tables	database/migrations/013_rls_policies.sql
IoT connectivity map	Simulated devices POST to REST API → verification engine	IS3-IOT-CONNECTIVITY-MAP.md , /api/iot/simulate
Analytics / Power BI	SQL views + REST analytics + filterable Chart.js dashboards	POWER-BI-SETUP.md , /api/analytics/*
AI workflow (academic)	Rule-based executive insights endpoint	IS3-AI-WORKFLOW.md , /api/analytics/ai-insights
Automated business rule (BPMN)	Daily expiry check job	src/jobs/expiryCheckJob.js

4.2 IP2 — Internet Programming 2

IP2 requirement	How DigiPermit addresses it	Evidence
RESTful API	Express MVC: routes → controllers → services	digipermit-backend/src/
Authentication & authorisation	Supabase Auth JWT + role middleware	/api/auth/*, roleMiddleware.js
CRUD operations	Full CRUD on organisations, users, FN, permits, alerts	Postman collection
Frontend integration	Ionic Angular consumes API with guards	digipermit-frontend/
API documentation	Open specification + Postman	API-SPECIFICATION.md
Testing	Test plan, role test guide, verify script	TEST-PLAN.md , ROLE-USER-STORIES-TEST-GUIDE.pdf

4.3 DS3 — Development Software 3

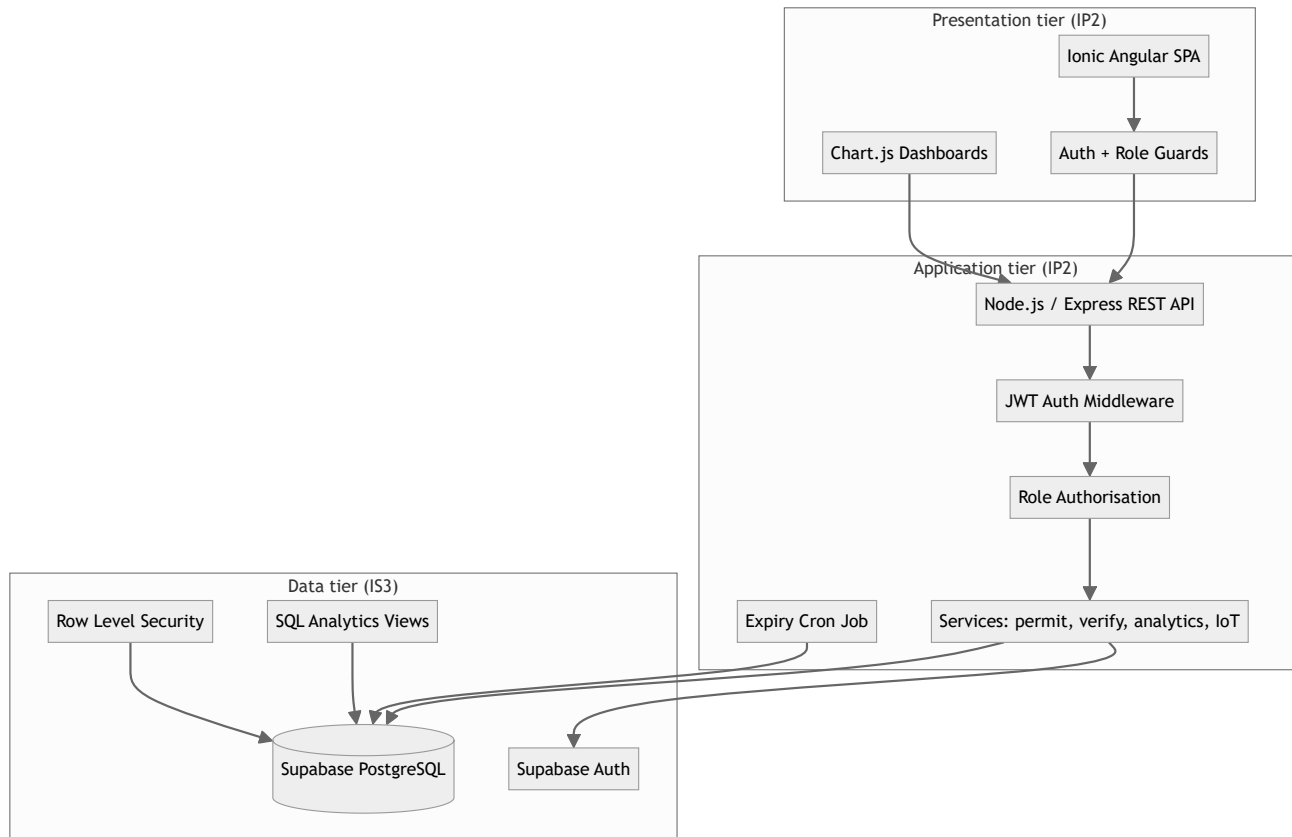
DS3 requirement	How DigiPermit addresses it	Evidence
Agile / Scrum	4 sprints with reviews and backlog	sprints/ , PHASE1-PLANNING-AND-DESIGN.md
Version control	GitHub repository with conventional commits	https://github.com/MrNtuli/DPermit
Working increments	Each sprint delivered testable features	Sprint review documents
Project documentation	Proposal, report, slides, demo script	PROJECT-PROPOSAL.md , FINAL-PROJECT-REPORT.md
Team collaboration	Shared repo, role-based demo walkthrough	DEMO-SCRIPT.md

4.4 Objectives mapping (project objectives O1–O7)

Objective	Implementation	Demonstration
O1 Centralise records	Multi-tenant PostgreSQL with org scoping	Admin + HR dashboards
O2 RBAC	8 roles, route guards, API <code>authorize()</code>	Negative test: HR cannot access <code>/admin</code>
O3 Expiry automation	Cron job + notifications	Seed permit near expiry; notifications list
O4 Multi-channel verification	Manual, QR, RFID endpoints + UI	Verification officer demo
O5 Suspicious activity	Rule-based alerts	Scan revoked permit; alerts page
O6 Analytics	SQL views, filtered charts, Power BI	Admin analytics filters
O7 End-to-end workflow	HR → FN → verify → immigration → manager	DEMO-SCRIPT.md

5. System architecture

5.1 Three-tier logical architecture

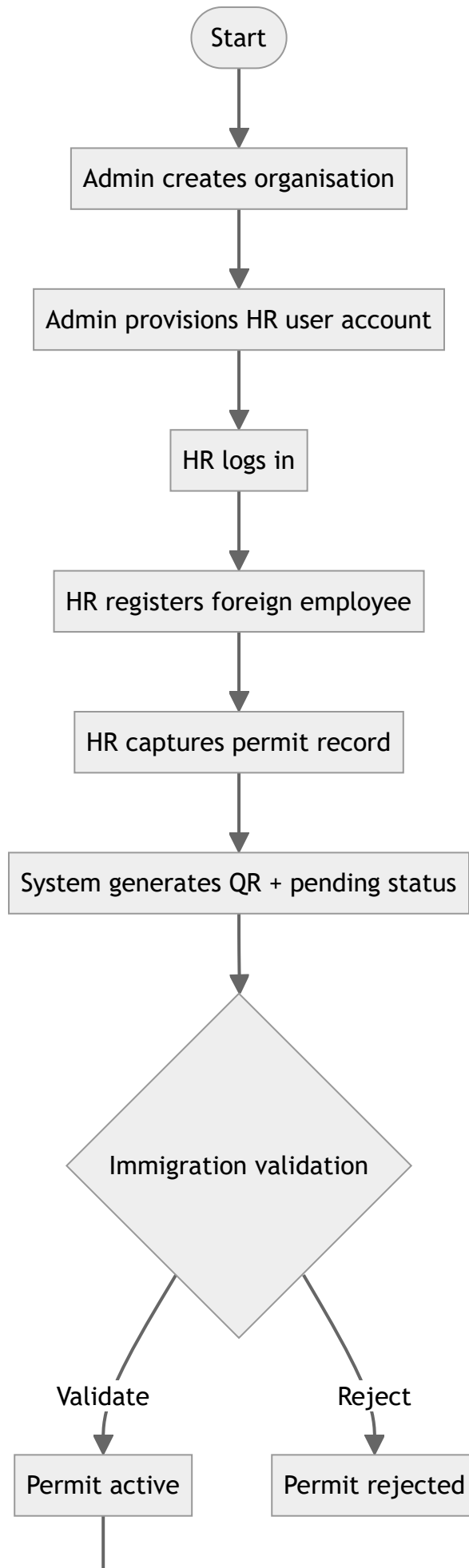


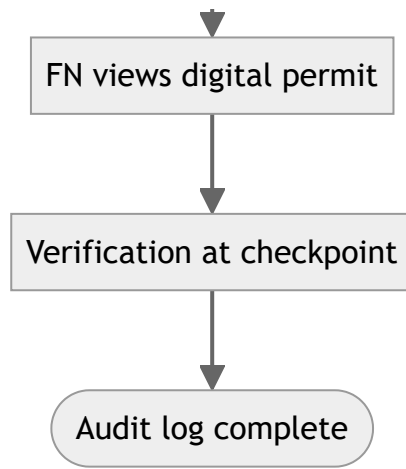
5.2 Technology stack

Layer	Technology	Purpose
Frontend	Ionic, Angular, TypeScript, SCSS	Mobile-first responsive UI
Charts	Chart.js	Live analytics dashboards
Backend	Node.js, Express.js	REST API, business logic
Database	Supabase PostgreSQL	3NF data store
Authentication	Supabase Auth (JWT)	Secure login
Hosting	GitHub Pages + Render	Live demo deployment
Analytics export	Power BI + SQL views	Executive reporting (IS3)
IoT demo	REST simulation + Packet Tracer map	Academic IoT integration
Testing	Postman, manual test guide	IP2 quality assurance
Source control	Git / GitHub	DS3 collaboration

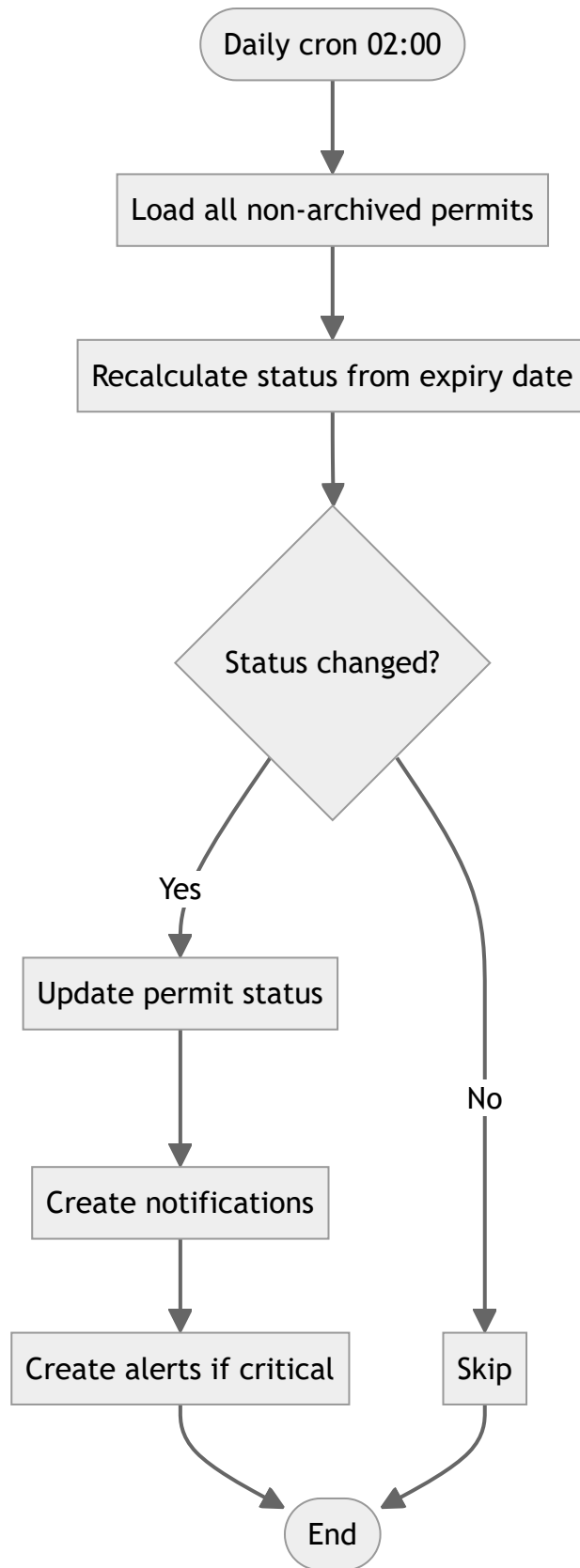
6. Process diagrams (BPMN-style)

6.1 UC-01 — Employer onboarding (core use case)

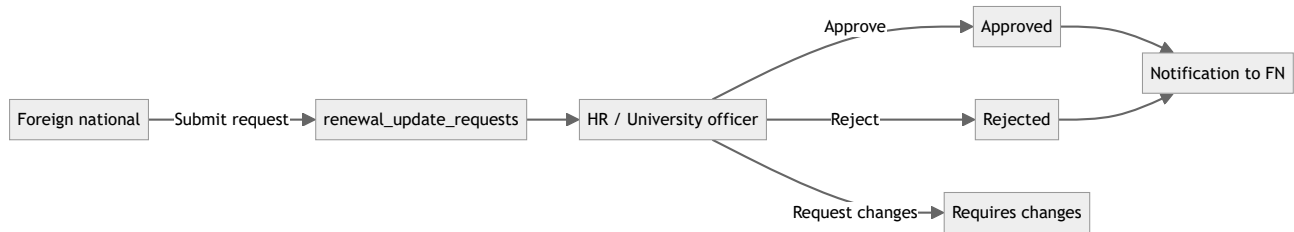




6.2 Expiry monitoring (automated BPMN activity)

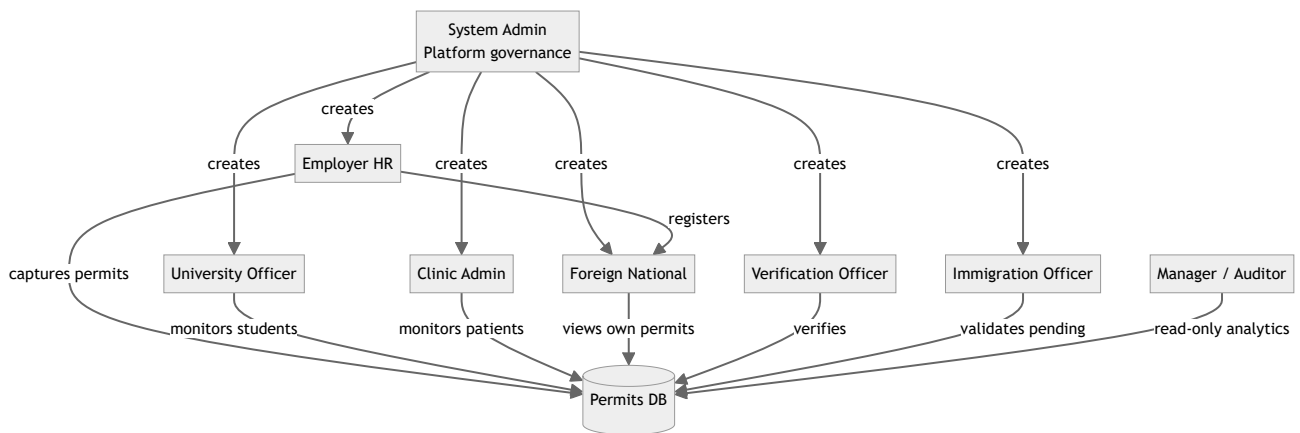


6.3 Renewal / update request flow



7. User roles and what happens for each

7.1 Role ecosystem

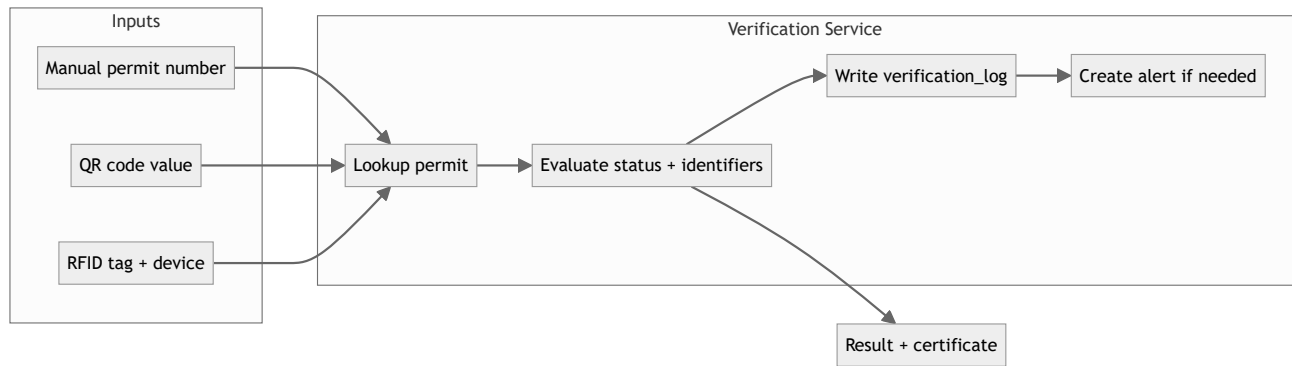


7.2 Role summary table

Role	Primary purpose	Key actions in system
System Administrator	Govern platform	Create orgs/users, view all data, analytics, logs, IoT devices
Employer HR	Work-visa compliance	Register employees, capture permits, verify (manual/QR), approve renewals
University Officer	Study-visa compliance	List students/visas, dashboard, verify (manual/QR)
Clinic Administrator	Patient document monitoring	List patients/permits, verify (manual/QR)
Foreign National	Self-service	View permits, QR document, notifications, request history
Verification Officer	Checkpoint	Manual, QR, RFID verification; certificates; logs
Immigration Officer	Simulated compliance review	Validate/reject pending permits, suspicious activity, alerts
Manager / Auditor	Executive oversight	Read-only analytics, AI insights report

8. Verification engine

8.1 Verification channels



8.2 Result states (examples)

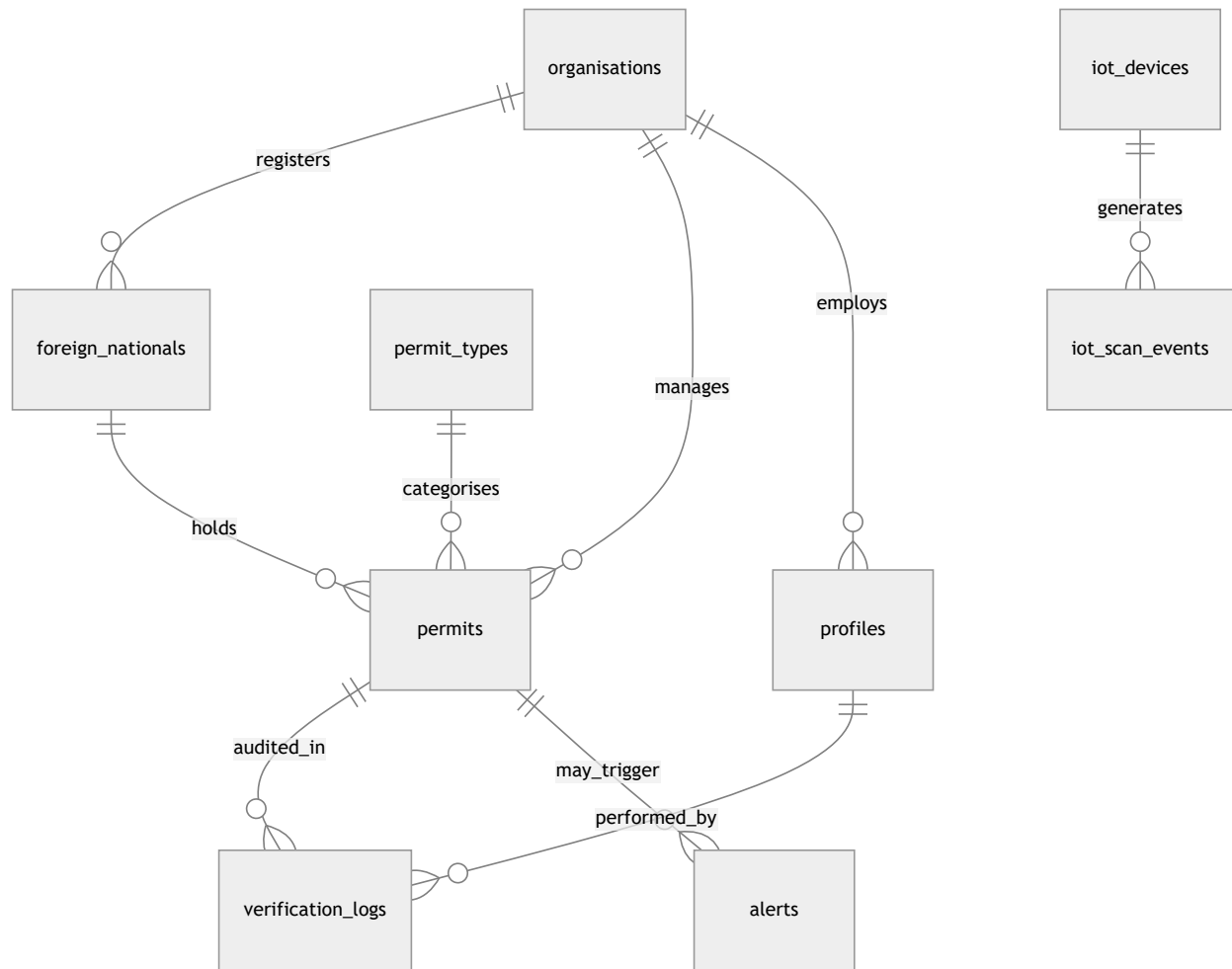
Result	Meaning	Typical trigger
valid	Active permit	Active record, before expiry
expiring_soon	Near expiry	Within threshold window
expired	Past expiry date	Expiry date passed
revoked	Administratively revoked	Status = revoked
not_found	No matching record	Unknown permit/QR
suspicious	Identifier mismatch	QR/RFID does not match record

8.3 Demo permit numbers

Permit number	QR value	Expected result
WP-2024-ACME-001	DIGIPERMIT:WP-2024-ACME-001	Valid
WP-2023-ACME-003	DIGIPERMIT:WP-2023-ACME-003	Expired
WP-2024-ACME-004	DIGIPERMIT:WP-2024-ACME-004	Revoked
SV-2024-METRO-001	DIGIPERMIT:SV-2024-METRO-001	Valid
MT-2024-CLINIC-001	DIGIPERMIT:MT-2024-CLINIC-001	Pending → validate via immigration

9. Data model (IS3)

9.1 Core entities (12 tables)



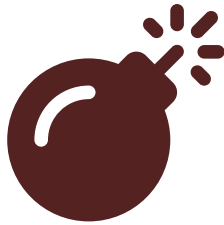
Full ERD: [IS3-ERD.md](#)

9.2 Multi-tenancy

- Each **organisation** (employer, university, clinic) owns its foreign nationals and permits.
 - HR/university/clinic users see **only their organisation** (enforced in API services + RLS).
 - Admin, manager, immigration (simulated) roles may have broader read access per role rules.
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10. Analytics and reporting (IS3)

10.1 Analytics data flow



Syntax error in text

mermaid version 10.9.6

10.2 Filterable analytics (lecturer requirement)

Users with analytics access can filter by:

- Organisation (admin/manager/immigration)
- Period (7 / 14 / 30 / 90 days)
- Scan type (manual / QR / RFID)
- Verification result
- Permit status
- Alert status

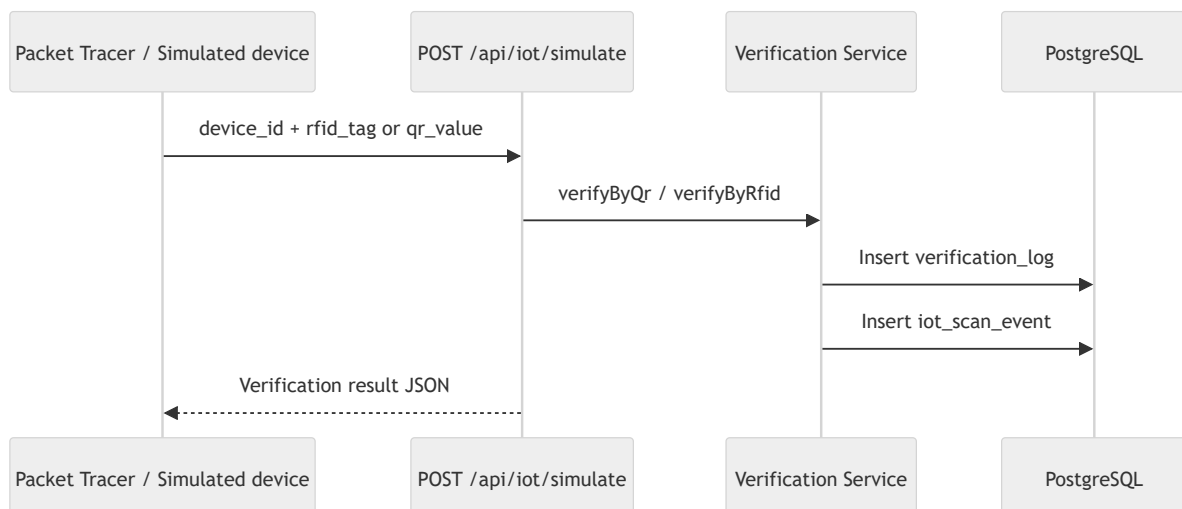
Filters apply to **charts and KPI cards** simultaneously via `/api/analytics/charts` and `/api/analytics/summary`.

10.3 Power BI

SQL views in Supabase connect to Power BI for executive dashboards.

Setup guide: [power-bi/POWER-BI-DASHBOARD-BUILD-GUIDE.md](https://github.com/immigrationnz/power-bi/blob/main/POWER-BI-DASHBOARD-BUILD-GUIDE.md)

11. IoT simulation (IS3)



Connectivity map: [IS3-IOT-CONNECTIVITY-MAP.md](#)

Lab guide: [PACKET-TRACER-LAB-GUIDE.md](#)

12. API layer (IP2)

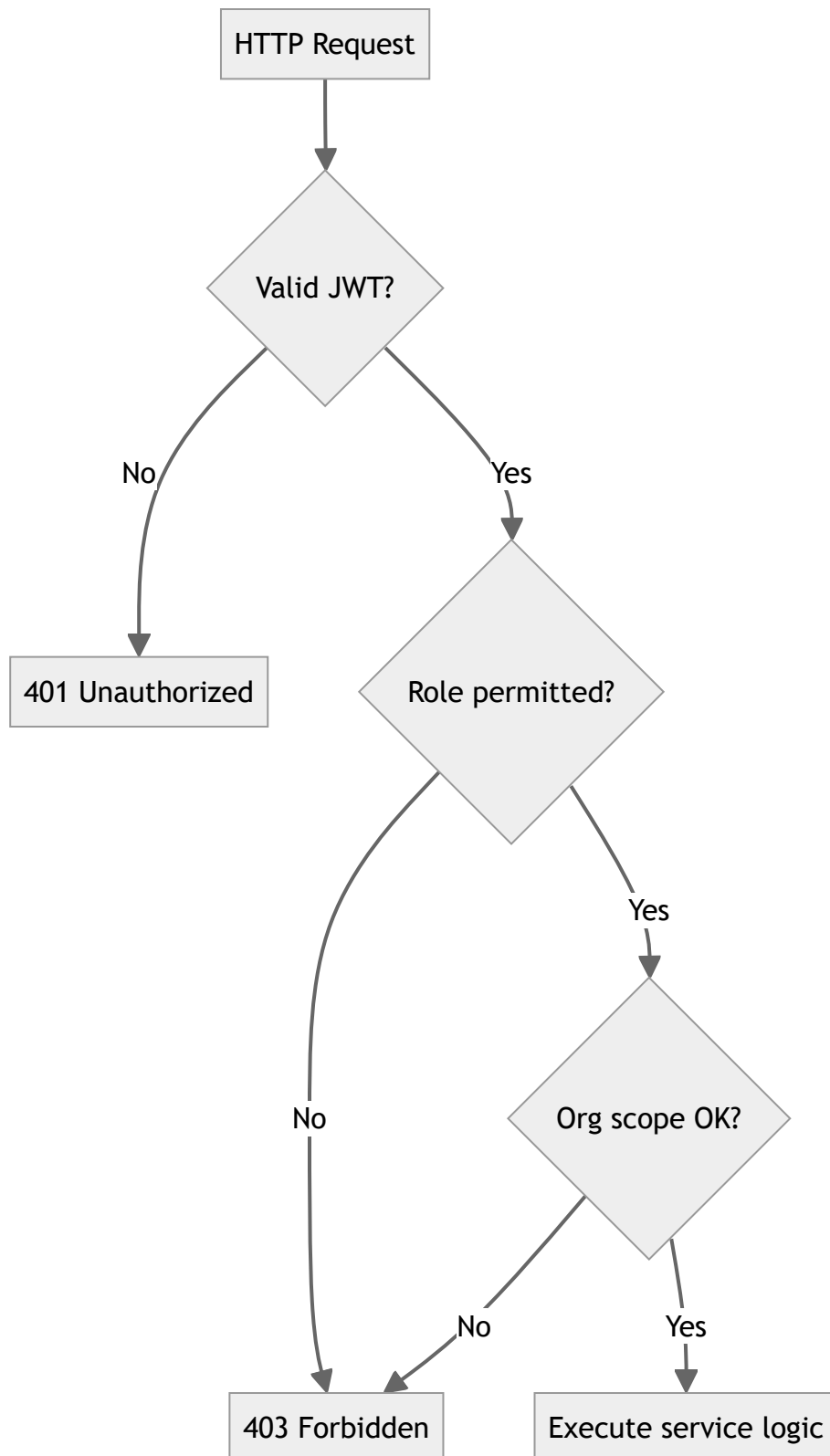
12.1 API structure

Group	Base path	Example endpoints
Auth	/api/auth	login, profile
Organisations	/api/organisations	CRUD (admin)
Users	/api/users	CRUD (admin)
Foreign nationals	/api/foreign-nationals	CRUD (org officers)
Permits	/api/permits	CRUD, validate, reject, revoke
Verification	/api/verify	manual, QR, RFID
IoT	/api/iot	simulate, devices, events
Analytics	/api/analytics	summary, charts, filters, ai-insights
Alerts / Notifications	/api/alerts , /api/notifications	list, resolve, mark read

Full specification: [API-SPECIFICATION.md](#)

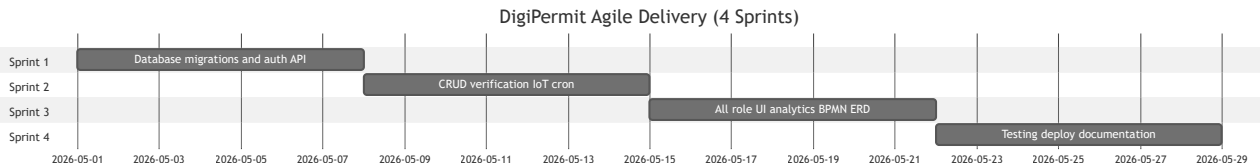
Postman: digipermit-backend/postman/DigiPermit-API.postman_collection.json

12.2 Security model



13. Agile delivery (DS3)

13.1 Sprint timeline



13.2 Increment delivered per sprint

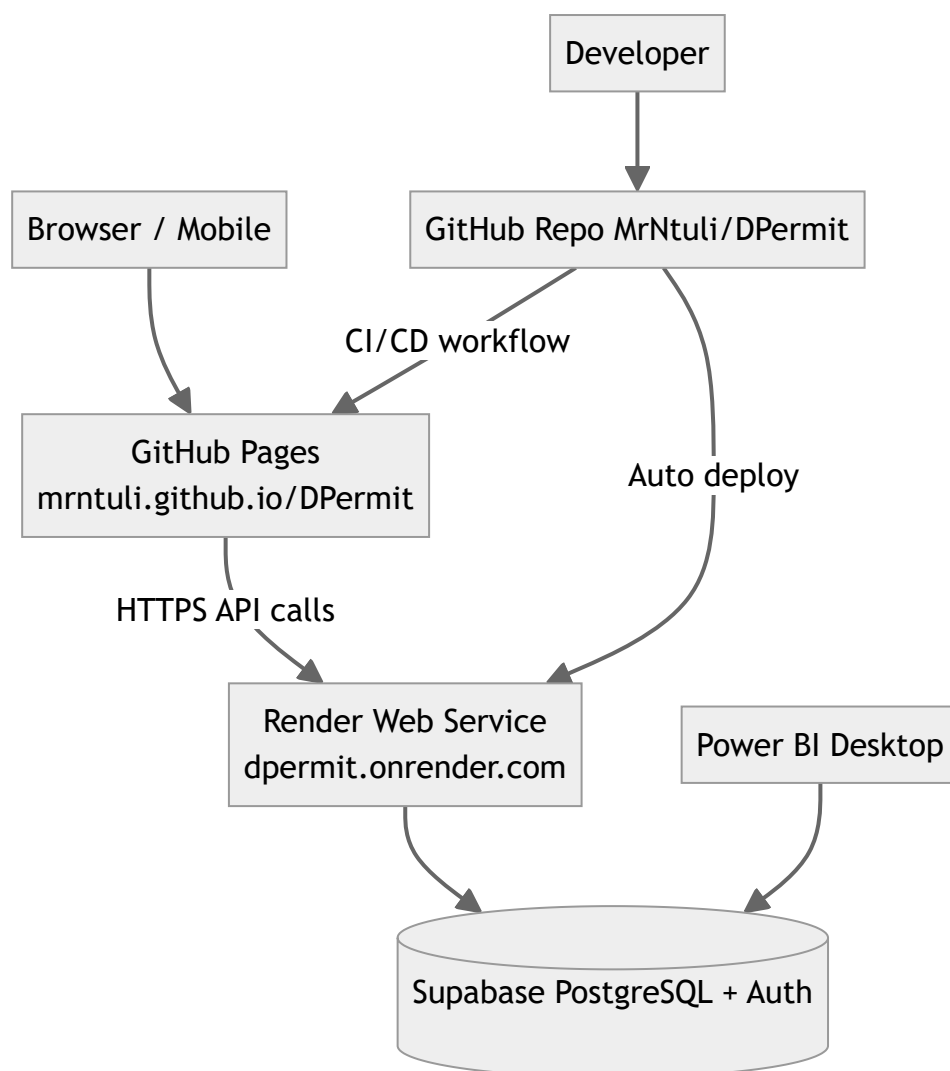
Sprint	Goal	Key deliverable
1	Backend & database foundation	12-table schema, auth, org/user API
2	Core business logic	Verification engine, IoT simulate, expiry job, HR UI
3	Full UI & IS3 artefacts	8 role dashboards, charts, BPMN/ERD/IoT docs
4	Release & presentation	Test guide PDF, live deploy, final report/slides

Reviews: [sprints/SPRINT-1-REVIEW.md](#) through [SPRINT-4-REVIEW.md](#)

14. Testing and quality assurance

Test type	Artefact	Purpose
API testing	Postman collection	IP2 endpoint verification
Role acceptance	ROLE-USER-STORIES-TEST-GUIDE.pdf	All 8 roles
System smoke test	<code>scripts/verify-system.ps1</code>	Health, login, permits, analytics
Live deployment	https://mrntuli.github.io/DPermit/	End-to-end demo
Test plan	TEST-PLAN.md	Formal IP2 test documentation

15. Deployment architecture



Component	URL
Source code	https://github.com/MrNtuli/DPermit
Live frontend	https://mrntuli.github.io/DPermit/
Live API	https://dpermit.onrender.com
API health	https://dpermit.onrender.com/api/health

Deploy guide: [DEPLOY.md](#)

16. Known limitations and future work

Limitation	Rationale	Future enhancement
Not official immigration issuer	Academic scope	N/A — by design
In-app notifications only	Phase 1 scope	Email/SMS integration
Simulated IoT / RFID	No hardware in capstone	Real device SDK
Admin-provisioned user accounts	Security model	HR invite flow
Supporting documents metadata only	No secure file vault	Document upload service
Rule-based AI insights	Academic AI layer	ML anomaly detection
Free Render tier sleep	Cost	Paid instance for production

17. Related documentation index

Document	Use
PHASE1-PLANNING-AND-DESIGN.md	Full requirements & FR/NFR
PROJECT-PROPOSAL.md	Week 1 proposal
FINAL-PROJECT-REPORT.md	10–15 page submission
PRESENTATION-SLIDE-DECK.md	Demo presentation
DEMO-SCRIPT.md	Live demonstration steps
COMPLETE-SETUP-GUIDE.md	Hosting checklist
CAPSTONE-DELIVERABLES-INDEX.md	All module deliverables

Document prepared for: Major Project submission — DS3 × IS3 × IP2

Team repository: <https://github.com/MrNtuli/DPermit>

PDF export: [pdf/MAJOR-PROJECT-SYSTEM-DESCRIPTION.pdf](#)

Regenerate: `python scripts/generate-system-description-pdf.py *`